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Maximizing Recovery from Marginal Reservoirs: A Full Cycle Development Approach Leveraging Best in Class Integrated Reservoir Management Practices

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Abstract

The field is an under saturated marginal oil reservoir with significant oil in place and has been producing since 2014. It has undergone an accelerated transition from green to brownfield with a notable increase in recovery factor (3%), while facing challenges such as pressure depletion and a high number of inactive strings. This paper presents the application of integrated reservoir management practices, focusing on reservoir health, dynamic monitoring, collaborative decision making, opportunity generation, target delivery, and digital dashboards to efficiently manage field maturation and enhance recovery across the

The methodology applied a dynamic synthesis approach leveraging integrated reservoir management (IRM) workflows to assess field challenges and develop targeted solutions. This included field performance reviews, sector level dynamic synthesis, and well level analysis. Key efforts focused on production optimization, recovery factor improvement by sector, and production injection rebalancing. Reservoir monitoring ensured compliance with pressure coverage and saturation logging guidelines, addressing challenges in deep oil bearing zones with limited vertical wells. Mitigations involved logging while tripping LWT deployment during well interventions. Opportunity generation, value of information analysis, and IRM dashboard implementation enabled integrated, data driven reservoir management.

The implementation of integrated reservoir management practices has led to significant improvements in field performance and reservoir health. Full alignment with the production mandate was achieved, delivering sustained output with a 30% recovery factor realized up to date. Water cut has been effectively maintained below 30%, and gas oil ratio (GOR) stability has been ensured. Reservoir pressure recovery has supported extended natural flow potential, enabled by optimized lower completion strategies and the implementation of hybrid artificial lift solutions. A notable reduction in inactive strings contributed to enhanced production efficiency and asset value. Dynamic synthesis and opportunity focused workflows facilitated the recovery of performance in previously underperforming areas, supporting full system optimization and production resilience.

Reservoir health management achievements include full compliance with volumetric replacement ratio (VRR) targets and consistent delivery of recovery expectations. Through rigorous surveillance, targeted

interventions, and digital integration via an IRM dashboard, the field successfully met both short term production assurance and long term technical rate targets. These results demonstrate the effectiveness of structured reservoir management workflows in overcoming marginal field challenges, maximizing resource recovery, and sustaining production throughout the maturation cycle.

The utilization of best reservoir management practices optimized recovery, minimized well inactivity, and enabled proactive decision making with digital tools, driving significant value and ensuring sustained field performance.

Introduction

The reservoir under study is a structurally complex, under saturated carbonate system located within a faulted anticline in a mature hydrocarbon province. It comprises two stratigraphic units deposited during the Barremian age and is part of the Thamama Group. The reservoir exhibits significant heterogeneity, with grain dominated facies in certain subzones and mud dominated facies in others. These geological characteristics contribute to variable permeability and porosity distributions across the field.

Since the commencement of full field production over a decade ago, the reservoir has transitioned from a greenfield to a brownfield asset. This maturation has been marked by a steady increase in recovery factor, now approaching double its initial value. Despite this progress, the reservoir has faced persistent challenges, including non uniform fluid movement, and a high proportion of inactive production strings due to high Water cut. Additionally, the limited number of vertical wells has constrained saturation monitoring, particularly in deeper oil bearing zones.

To address these challenges, a comprehensive reservoir management strategy was implemented. This strategy emphasized dynamic synthesis, sector level diagnostics, and well level analysis. Key interventions included production injection rebalancing, artificial lift optimization, and targeted well interventions. The approach was underpinned by integrated reservoir management (IRM) workflows designed to maximize recovery and sustain production across the asset lifecycle.

Methodology

Reservoir Health Monitoring and Diagnostics

Reservoir health assurance was conducted through a structured, multi sector optimization strategy aimed at sustaining the Field Sustainable Operational Production Rate (FSOPR) and maximizing recovery. This approach integrated diagnostics, surveillance, and targeted interventions to maintain reservoir integrity and production resilience.

Production optimization was executed across all sectors, tailored to the specific performance dynamics of each region. In the central sector, strategic relaxation of production rates enabled pressure recovery, reversing prior depletion trends and restoring reservoir energy [Figure 1](#). The northern sector demonstrated measurable improvement in recovery factor through optimized completions and targeted well interventions. In the southern sector, which had previously exhibited performance degradation, production was successfully restored through sidetracking, artificial lift enhancements, and injection redistribution.

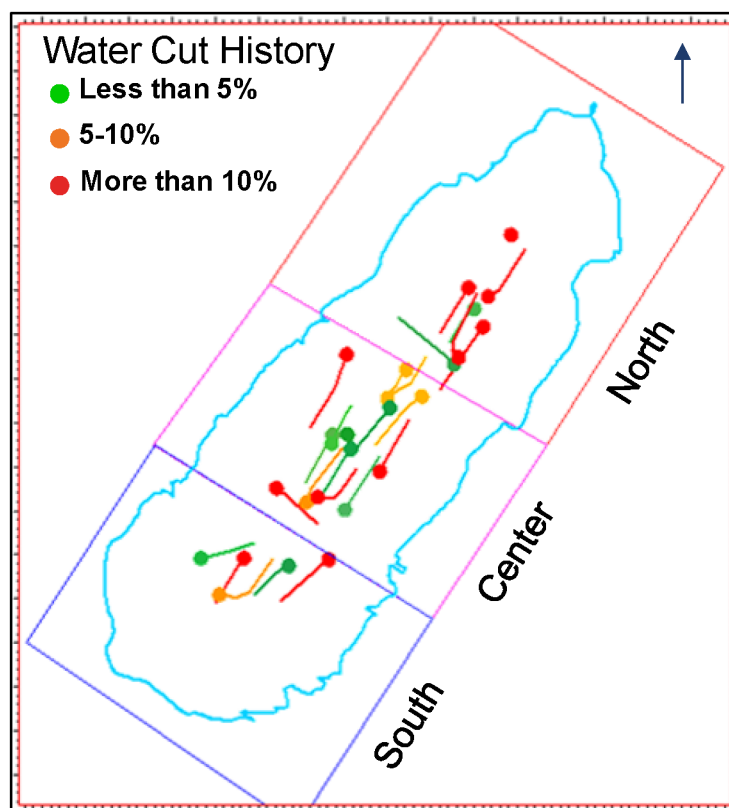


Figure 1—Reservoir Map showing the Sectors: North, Central and South

A key outcome of this reservoir health strategy was the significant reduction in inactive strings. Through selective workovers and reactivation campaigns, well availability was improved, directly contributing to the achievement of the FSOPR quota for the last 4 years. This operational milestone was supported by full compliance with the volumetric replacement ratio (VRR), with 100% target attainment recorded for the same period.

Cumulatively, these efforts resulted in a realized recovery factor of approximately 30%, reflecting a consistent annual improvement rate of ~3%. This performance underscores the effectiveness of the reservoir health management framework in sustaining production, enhancing recovery, and ensuring long term field viability.

Surveillance Techniques and Data Integration

A robust surveillance program was established to support proactive reservoir management and ensure excellent pressure and saturation coverage across the field. This program integrated historical datasets with real time monitoring tools, enabling continuous diagnostics and performance tracking. Data acquisition included modular formation dynamics tester (MDT), pressure volume temperature (PVT) sampling, production logging tools (PLT), and reservoir saturation tools (RST), with over 60 surveillance jobs executed across the reservoir. These datasets were visualized through digital dashboards, facilitating rapid interpretation and informed decision making.

Continuous monitoring of pressure trends, saturation profiles, and well performance metrics was central to the surveillance strategy. Pressure build-up (PBU) and fall-off (PFO) surveys were conducted across multiple sectors to evaluate reservoir connectivity, pressure support, and depletion behavior. Saturation mapping was performed using a combination of open-hole logs, PLT data, and newly acquired (LWT) measurements. The LWT technique was deployed during well interventions to overcome the limitations of

vertical well coverage, particularly in deeper oil-bearing zones. These diagnostics enabled the identification of swept zones, high water cut intervals, and underperforming wells requiring intervention.

The reservoir was subdivided into stratigraphic subzones using high-resolution sequence stratigraphy. This allowed for targeted analysis of lithofacies and permeability contrasts, which were critical for optimizing development strategies. Subzones characterized by grain-dominated facies, such as rudist-peloid rudstone (RPR) and skeletal peloid grainstone (SPG) were prioritized due to their superior reservoir quality and flow potential. Diagnostic plots and conformance maps were employed to correlate water cut evolution with recovery trends, guiding the selection of candidate wells for water shut-off and sidetracking.

Pressure maps were updated annually to track reservoir depletion and pressure recovery dynamics. Saturation maps were refined using LWT and open-hole log data, particularly in areas where traditional saturation logging was constrained. Surveillance also encompassed well performance tracking, injection profiling, and artificial lift diagnostics. This multi-source data integration enabled timely interventions, optimized reservoir sweep, and sustained production performance across the asset lifecycle. [Figure 2](#)

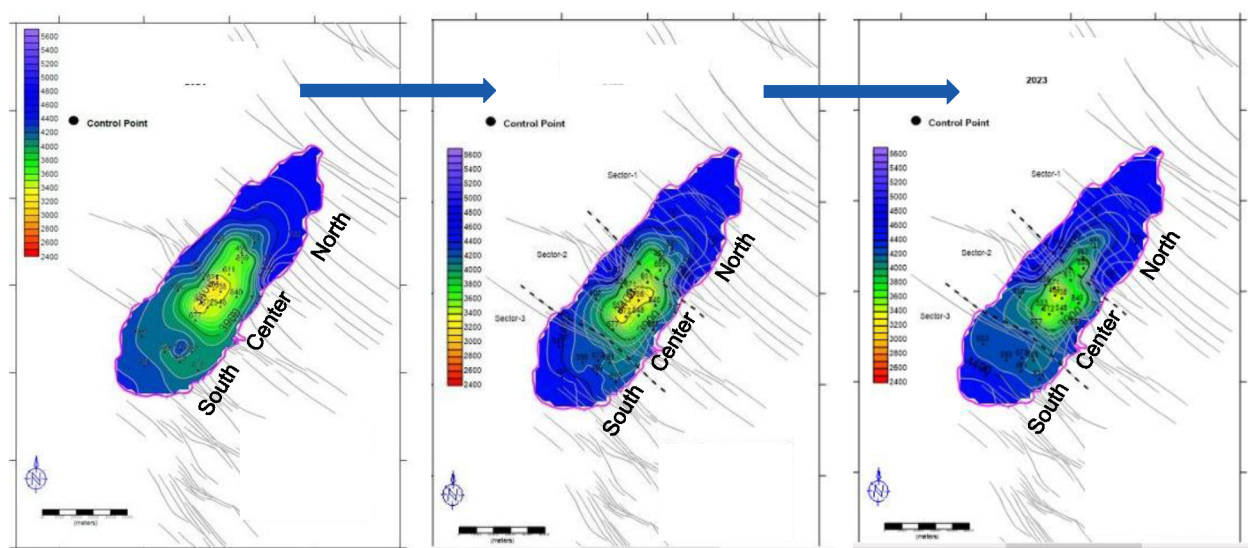


Figure 2—Reservoir isobaric maps over the past three years illustrating the incremental changes across various sectors

Cross-Disciplinary Collaboration and Opportunity Generation

Opportunity generation was enabled through close collaboration among subsurface, production, and data science teams. Regular performance reviews and Sector Performance Review (SPR) workshops facilitated alignment across disciplines, allowing for integrated evaluation of reservoir behavior and operational constraints.

Key interventions included sidetracks and infill drilling in high-potential zones located in the bottom part of the reservoir, guided by saturation. These efforts yielded significant production gains and avoided the cost of new well drilling. Value of Information (VOI) analysis supported prioritization of opportunities based on technical and economic impact.

Water shut-off (WSO) operations were executed in wells with high water cut. Artificial lift strategies were optimized through phased gas lift tie-ins and the deployment of electrical submersible pumps (ESP) in wells requiring enhanced lift capacity.

Lower completion designs were improved with the use of inflow control devices (ICDs), which helped manage inflow distribution and reduce water production. These completions, combined with targeted diagnostics and collaborative workflows, enabled recovery of performance in previously underperforming areas and supported sustained field output.

Target Delivery Strategies

Production targets were met through strategic rebalancing of injection and offtake across reservoir sectors, tailored to sector-specific characteristics and performance trends. This approach ensured pressure support in depleted zones and optimized sweep efficiency in high-potential areas.

The implementation of key opportunities: including sidetracks, infill drilling, water shut-off (WSO), artificial lift via gas lift and ESP systems, and the deployment of lower completions with inflow control devices (ICDs), was instrumental in stabilizing production and enhancing reservoir performance.

These interventions enabled the field to maintain its production plateau to date, with a recovery factor reaching 30% over a 10-year period. Water cut has been consistently managed below 30%, supporting sustained oil productivity and reservoir health.

This outcome reflects the effectiveness of targeted delivery strategies integrated within the broader reservoir management framework, ensuring alignment with operational mandates and long-term recovery objectives.

Digitalisation and Automation Tools

Digital tools played a pivotal role in enhancing reservoir management efficiency. Integrated dashboards were developed to support workflows for the Major Field Review (MFR), Sector Performance Review (SPR), and Integrated Reservoir Management (IRM). All data were organized into the automated SPR workflow through a web based Business Process Management (BPM) that provided mechanism for the user to load, validate and approve technical data (Martinez Santiago, Al Shehhi, Anbalagan, & Shbair, 2022), ensuring consistency and traceability across the reservoir management lifecycle.

Real-time visibility into key performance indicators KPI (Martinez Santiago, Al Shehhi, Anbalagan, & Shbair, 2022) was achieved through these dashboards, enabling data-driven decision-making and rapid response to evolving reservoir conditions. The SPR dashboard served as the central interface for performance tracking, while the IRM dashboard supported strategic planning and execution.

Opportunity identification was streamlined using the IRM workflow-based Opportunity Dashboard, specifically the IRM workflow 8 WF.8 Opportunity Management module (Martinez Santiago, Al Shehhi, Anbalagan, & Shbair, 2022). This tool captured outcomes from sector performance reviews and summarized them in a centralized opportunity register. It facilitated effective opportunity maturation, detailed planning, and execution tracking, ensuring alignment with reservoir development objectives.

The implementation of automated SPR workflows and digital dashboards has transformed traditional work processes into advanced, analytics-driven systems.

Integrated Reservoir Management Workflows

Petroleum Industry is dedicating a great effort in the standardization and automation of operational business processes. As part of the digital transformation, ADNOC (Abu Dhabi National Oil Company) has embraced a framework named Integrated Reservoir Management (IRM) as part of key management best practices (Al Marzouqi, Bahamaish, Al Jenaibi, Al Hammadi, & Saputelli, 2016).

Integrated Reservoir Management (IRM) is defined as a structured framework of interlinked building blocks, processes, and workflows that support reservoir management objectives from surveillance strategy to opportunity execution. It provides a standardized approach for managing subsurface resources, aligning technical practices with corporate goals, and promoting consistency across operating assets. The IRM framework enables multidisciplinary integration, promotes best practices, and ensures governance through clearly defined roles, responsibilities, and performance indicators. (Al Marzouqi, Bahamaish, Al Jenaibi, Al Hammadi, & Saputelli, 2016)

The framework is applied at multiple levels: field, sector, and well, through systematic performance reviews. At the field level, IRM facilitates strategic planning and continuous monitoring of reservoir health, production sustainability, and recovery optimization. Sector level implementation focuses on dynamic

synthesis and opportunity identification, while well level diagnostics guide targeted interventions such as artificial lift deployment, water shut-off, and completion optimization. These layers of analysis are supported by standardized workflows that ensure repeatability, traceability, and alignment with business objectives.

IRM also fosters a collaborative environment by integrating data acquisition, validation, visualization, and opportunity generation into a continuous improvement cycle. Opportunities identified during sector and well performance reviews are captured in an opportunity register and matured through technical evaluation. The framework's emphasis on automation, digital dashboards, and workflow standardization has transformed traditional reservoir management into a data-driven, value-focused discipline that supports sustained production and enhanced recovery.

Results

Recovery Improvement and Production Optimization

Significant recovery improvements were achieved through targeted interventions and opportunity maturation. Two sidetracked wells in high-potential zones successfully reduced water cut and delivered oil gains exceeding 60% per well. Additionally, two wells were equipped with electrical submersible pumps (ESP), and one water shut-off (WSO) operation was performed using lower completion strategies ICD. These actions contributed to a production gain of 5 MBOPD, representing 30% of total field output. The southern sector, previously underperforming, increased its contribution by more than 30%, reflecting the effectiveness of sector-level optimization.

The reservoir delivery targets were fully met, including maintaining a 20% technical rate (TR), achieving 100% compliance with the production mandate, and restoring output from the southern area. There is a significant reduction in inactive strings leading to current recovery factor (RF) stands at 30% Figure 3, with an expected ultimate recovery (EUR) of 53%. Enhancements to the dynamic model will extend the production plateau by four years, now projected to last until Q1 2028.

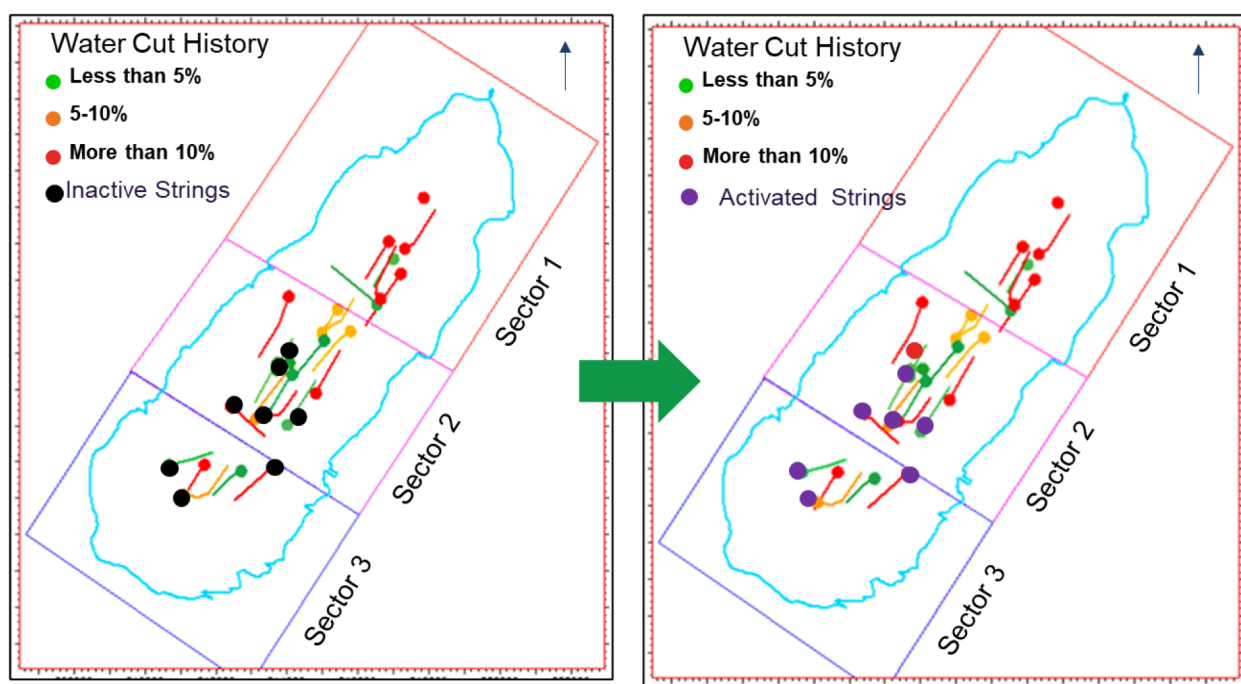


Figure 3—Reservoir Map displaying the Inactive String evolution.

Cost Reduction and Efficiency Gains

A proactive strategy was adopted to mitigate delays in gas lift tie-in packages and ensure production continuity. ESPs were introduced as backup lift mechanisms, and sidetracks were executed to enhance natural flow potential. These sidetracks avoided the need for three new infill wells, resulting in a cost saving. Logging While Tripping (LWT), Production Logging Tools (PLT), and upper reservoirs deepening supported precise opportunity identification, enabling efficient execution and minimizing operational overhead.

Impact of Methodology and Workflows

The integrated reservoir management framework enabled systematic opportunity generation and execution. Sector performance reviews and well level diagnostics guided interventions such as sidetracking, WSO, and artificial lift deployment. Opportunities were captured in a centralized register and matured through technical evaluation. The use of digital dashboards and automated workflows ensured real time visibility, streamlined decision making, and alignment with reservoir development objectives. These structured processes directly contributed to enhanced recovery, reduced costs, and sustained production performance.

Conclusion

This study demonstrates that marginal reservoirs can achieve substantial recovery and sustained performance through structured and integrated reservoir management. The implementation of dynamic synthesis, sector level diagnostics, and well level optimization enabled a transition from green to brown field status, achieving a recovery factor of 30% over 10 years, while maintaining a water cut below 30% and ensuring pressure recovery.

Key learnings from this development include:

- Continuous reservoir health monitoring and diagnostics were critical in identifying non uniform fluid movement, pressure depletion, and inactive strings, enabling timely and targeted interventions.
- Surveillance integration and digital tools, including automate SPR, and IRM workflows, enhanced decision making and operational efficiency through real time dashboards and performance tracking.
- Cross disciplinary collaboration facilitated opportunity generation, such as sidetracks and infill optimizations, and ensured alignment across reservoir, production, and operations teams.
- Strategic production injection rebalancing and artificial lift optimization, including hybrid lift systems and lower completion strategies, supported sustained output and extended natural flow potential.
- Sidetracking and selective completions provided cost effective solutions for reactivating inactive strings, contributing to a production gain of 5 MBOPD equivalent to 30 % of the total output and saving over 13.5 million USD in drilling costs.

The approach outlined in this paper is applicable to other marginal reservoirs and mature fields facing similar challenges. By adopting integrated workflows, leveraging digital tools, and fostering collaboration, operators can maximize recovery, enhance efficiency, and ensure long term asset value.

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the integrated reservoir management strategies discussed in this paper. Their commitment to excellence and crossdisciplinary collaboration was vital to the success of this project.

Nomenclature

IRM	Integrated Reservoir Management
RF	Recovery Factor
EUR	Estimated Ultimate Recovery
FSOPR	Field Sustainable Operational Production Rate
TR	Technical Rate
GOR	Gas Oil Ratio
WC	Water Cut
VRR	Volumetric Replacement Ratio
ESP	Electrical Submersible Pump
WSO	Water Shut-Off
ST	Sidetrack
PLT	Production Logging Tool
LWT	Logging While Tripping
MDT	Modular Formation Dynamics Tester
PVT	Pressure Volume Temperature
PBU	Pressure Build Up
PFO	Pressure Fall Off
KH-1	Specific well identifier used in deepening and data acquisition
SPR	Sector Performance Review
MFR	Major Field Review
BPM	Business Process Management
WF.8	IRM Workflow 8: Opportunity Management
Gate 9	Development planning milestone used in dynamic modeling
OPCO	Operating Company
IOC	International Oil Company
NOC	National Oil Company
DOF	Digital Oilfield
RPR	Rudist-Peloid Rudstone
SPG	Skeletal Peloid Grainstone
ICD	Inflow Control Device
SW	Water Saturation
HC	Hydrocarbon
MBOPD	Thousand Barrels of Oil Per Day

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